Lecture 5: Sep 15, 2026

Last time:

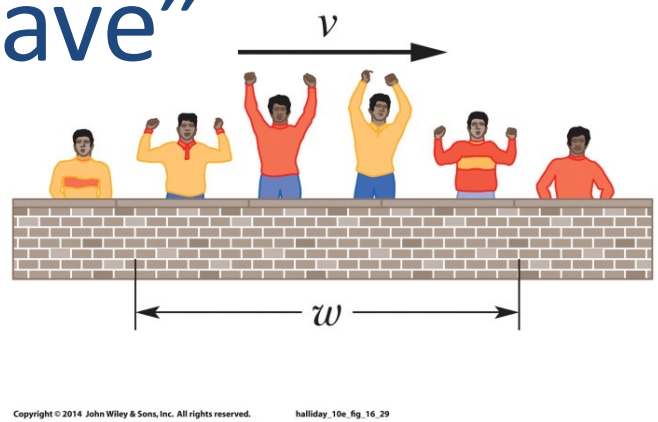
- Review of complex math
- Phasor representation of SHM, using phasors

Today's class:

- Transverse vs longitudinal waves
- Travelling waves and the wave equation
- Total phase for sinusoidal waves
- Superposition of waves

WAVES

"The wave"



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http://sociologyinfocus.com/tag/sports/

Actually a pretty good way to think about wave motion

- Each person does the same as the person beside them, but at a short time Δt later

The "disturbance" travels at a speed v :

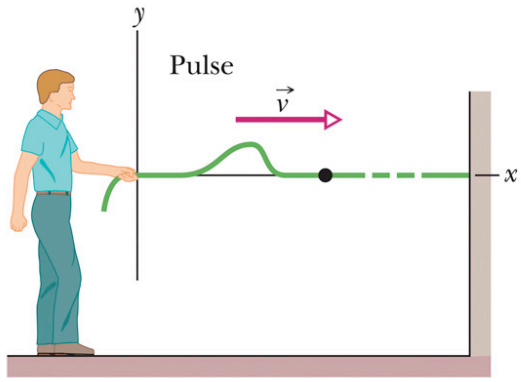
$$v = \frac{d}{\Delta t}$$

with Δt the time lag and d the distance from one seat to the next

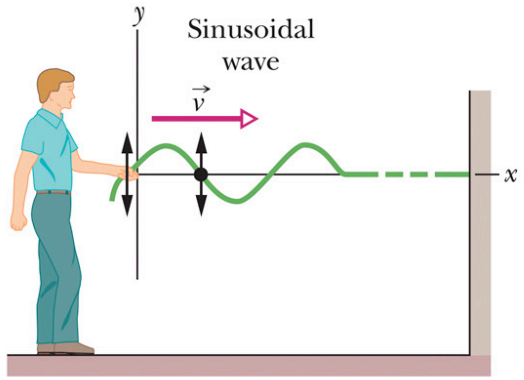
Longitudinal and transverse waves

A wave on a string is an example of a **transverse wave**.

As the wave passes, each particle of the string moves up and down, **transversely** to the motion of the wave itself.



(a)



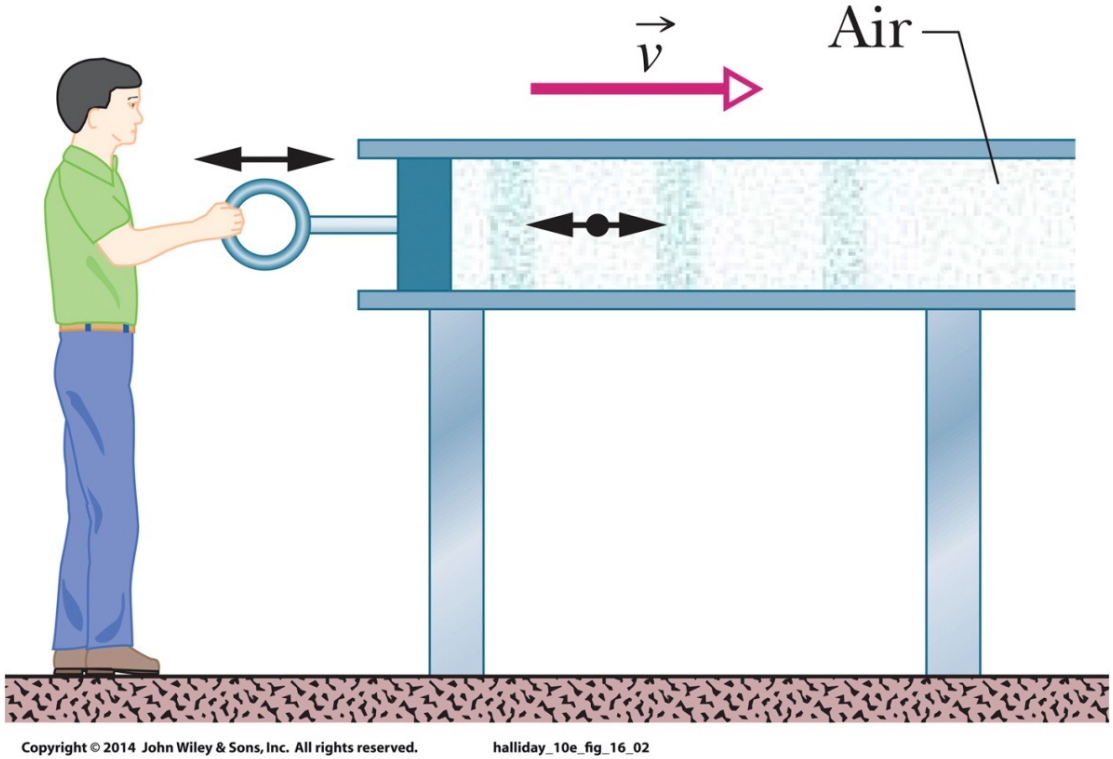
(b)

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Longitudinal and transverse waves

A sound wave is an example of a **longitudinal wave**.

As the wave passes, each particle of air moves forward and back, **parallel** to the motion of the wave itself.



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Compression Waves in Traffic

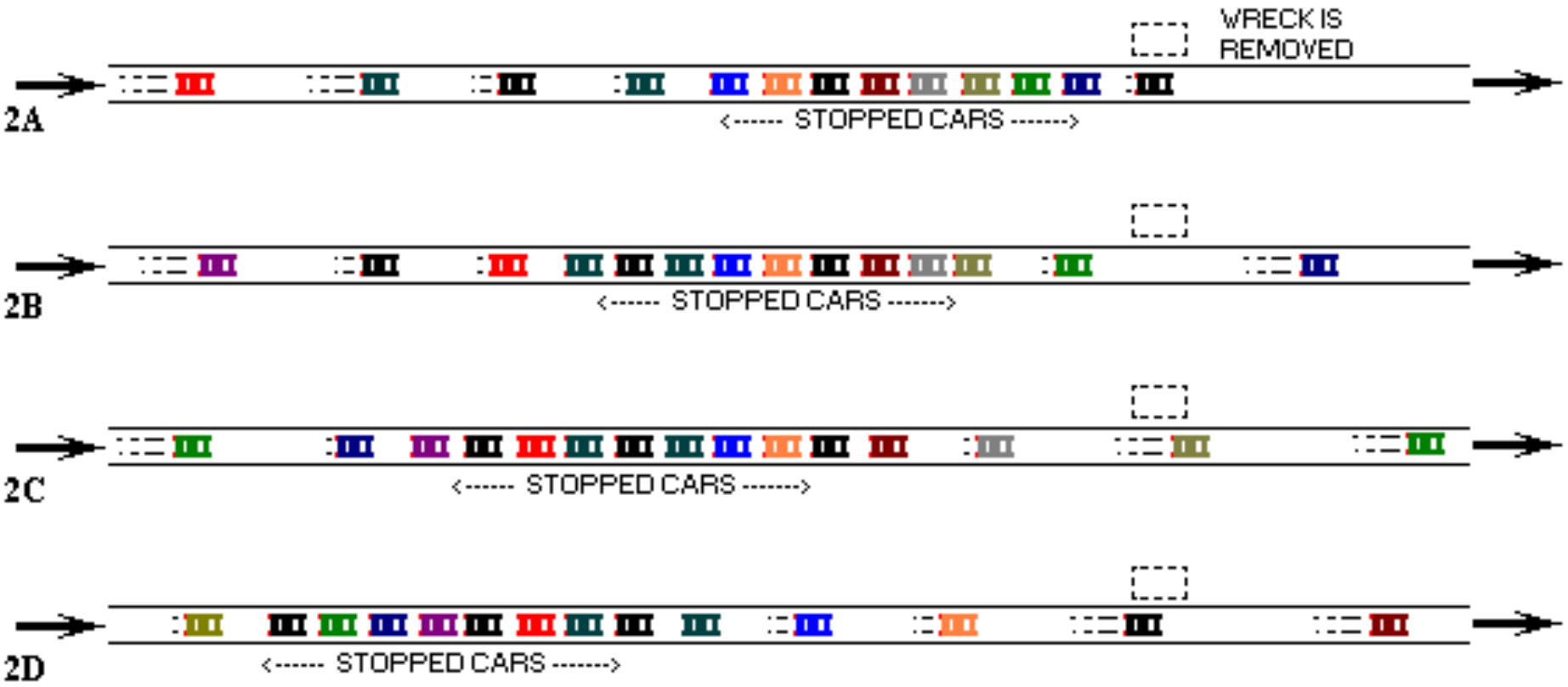


Image source: http://trafficwaves.org

Travelling waves

A fixed pattern ("wave function") moving at a velocity v ("phase velocity") is written like this:

y = wave function:
- hand height
- string height
- pressure
- water height
- …

$$y(x,t) = f(x \pm vt)$$

+: moves to the left
-: moves to the right

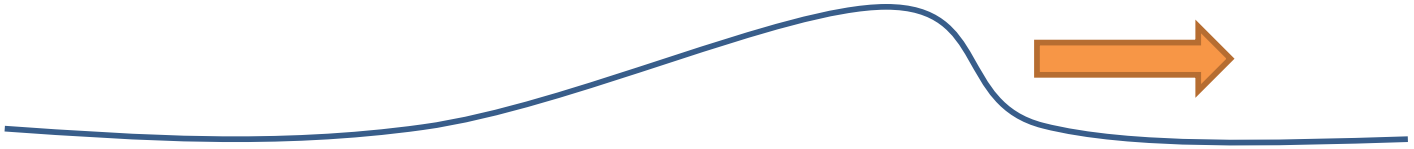
f is **ANY** single-valued function where x and t appear only in the combination $x \pm vt$ (with v constant)

At this stage, nothing else is special about f.

y is a function of 2 variables (x,t)

The wave equation

Acceleration and spatial structure are linked.



Any wave function $y(x,t) = f(x \pm vt)$ is a solution of the **wave equation**:

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$$

Where v is the wave speed

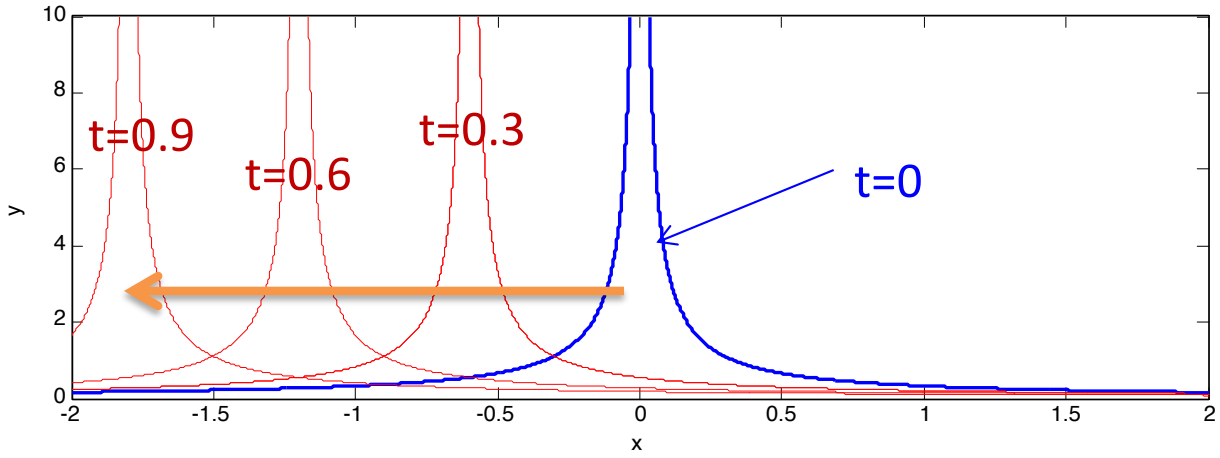
Example

$$y(x,t) = \left|\frac{1}{-3x-6t}\right|$$

Can be written

$$y(x,t) = \frac{1}{3}\left|\frac{1}{x+2t}\right| = f(x \pm vt)$$

and is therefore a solution to the wave equation with $v = 2$. Since v>0, the pattern moves to the left.



Sinusoidal waves

A sinusoidal wave is a special case where the function $\mathrm{f}(\mathrm{x} \pm vt)$ is a sinusoid:

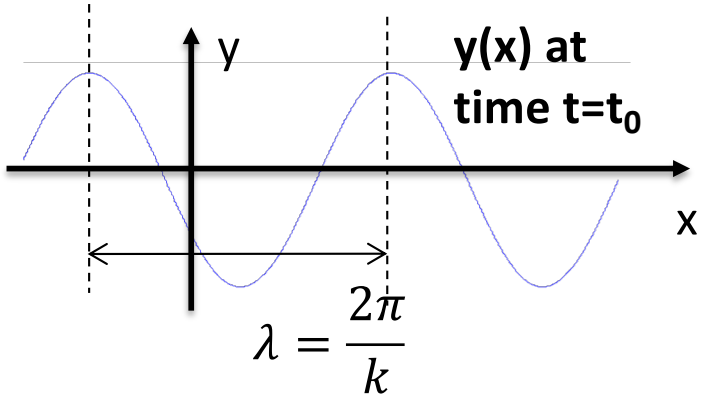
$$y(x,t) = A\cos(kx \pm \omega t + \phi_0)$$

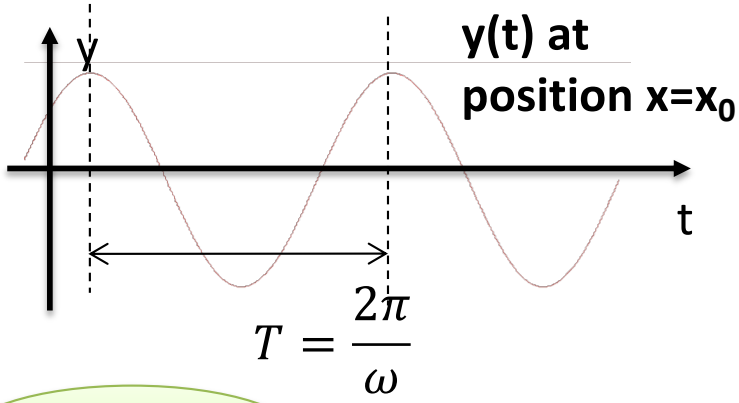
- Initial phase
- Amplitude
- Angular wavenumber $k = 2\pi/\lambda$
- Angular frequency $\omega = 2\pi f$

The wave moves at a speed ("phase speed")

$$v = \frac{\omega}{k} = f\lambda$$

A note on "phase" (or "total phase")





$$y(x,t) = A\cos(kx - \omega t + \phi_0)$$

- The quantity $kx - \omega t + \phi_0$ is the (total) wave phase
  - Argument to sin/cos
  - ϕ_0 is the initial phase at t=0,x=0
- If graphed as a function of position, the total wave phase changes by 2π per wavelength
- If graphed as a function of time, the total wave phase changes by 2π per period

Solving for λ:

$$\lambda = v/f = (344 \text{ m/s}) / 1000 \text{ Hz} = 34.4 \text{ cm}$$

So that:

$$|\Delta\phi| = 2\pi\left(\frac{8\ cm}{34.4\ cm}\right) = 1.46 \text{ rad} \quad \text{or} \quad 83.7^o$$

Note that the sign of $\Delta\phi$ is unknown, and depends on the direction in which the wave travels.

HARDER EXAMPLE: The same wave (v=344 m/s, 1000 Hz) passes 2 observers. The displacement at $x = 0$ is

$$y(t, x = 0) = A\cos(\omega t)$$

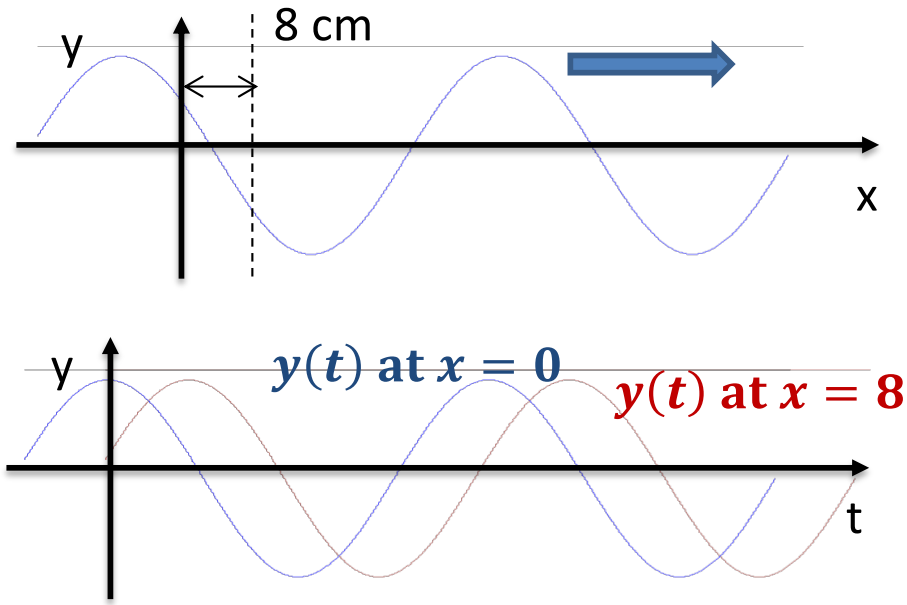
If the wave is travelling to the right, what is $y(t)$ at $x = +8$ cm?

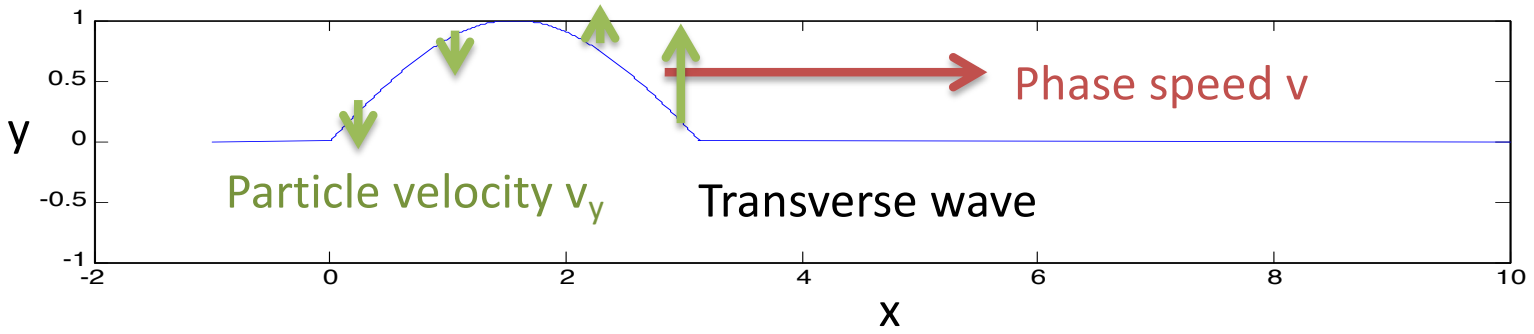
From the last problem:

$$|\Delta\phi| = 1.46 \text{ rad}$$

If the wave moves to the right, then the crest passes x=8 cm **after** it passes x=0 (i.e. the sinusoid shifts to the right). Therefore, choose $\Delta\phi < 0$ and so

$$y(t, \mathrm{x} = 8\mathrm{cm}) = \mathrm{A}\cos(\omega t - 1.46)$$





Phase (wave) velocity vs. particle velocity

- Phase velocity (wave velocity) is the velocity at which the wave propagates
- Use the symbol v
- In the x direction above
- Constant – not a function of position

- Particle velocity (medium velocity) is the velocity of the particles in the medium
- Use the symbol v_y or v_s
- Transverse wave: in the y direction

$$v_y = \frac{\partial y(x,t)}{\partial t}$$

- Longitudinal wave: in the same direction as the wave (x)
- Function of position – not constant

Velocities for sinusoidal waves

For a sinusoidal wave

$$y(x,t) = A\cos(kx \pm \omega t + \phi_0)$$

the velocity of the particles is

$$v_y = \frac{\partial y(x,t)}{\partial t} = \mp A\omega \sin(kx \pm \omega t + \phi_0)$$

While the velocity of the wave is

$$v = \frac{\omega}{k}$$

SUPERPOSITION

Wave superposition

If $y_1(x,t)$ and $y_2(x,t)$ are both solutions to the wave equation then:

$$y_3(x,t) = y_1(x,t) + y_2(x,t)$$

is also a solution to the wave equation.



Image: http://cnx.org/contents/031da8d3-b525-429c-80cf-6c8ed997733a@8.8:126/College_Physics

→Waves can pass through each other

→Waves can be constructed in linear combinations

Proof

more info

If $y_1(x,t)$ and $y_2(x,t)$ are both solutions to the wave equations, then they can be written as :

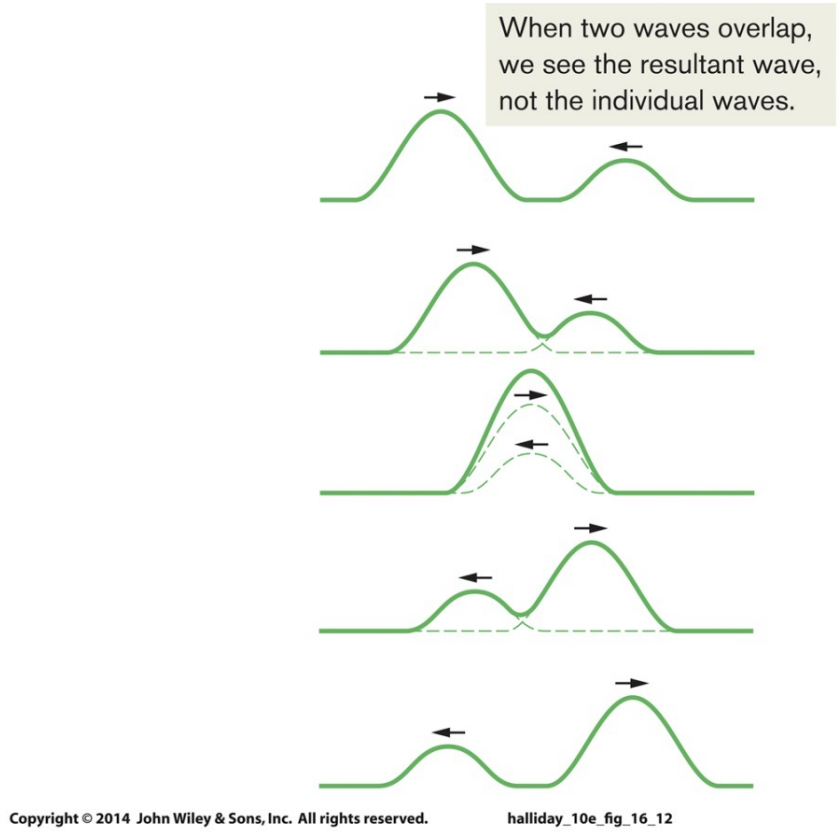
$$\begin{cases} y_1(x,t) = f_1(\mathrm{x} \pm vt) \\ y_2(x,t) = f_2(\mathrm{x} \pm vt) \end{cases}$$

so that

$$\begin{aligned} y_3(x,t) &= y_1(x,t) + y_2(x,t) \\ &= f_1(\mathrm{x} \pm vt) + f_2(\mathrm{x} \pm vt) \end{aligned}$$

is also a function of $\mathrm{x} \pm vt$ and therefore also a solution to the wave equation.

Examples: waves on a string



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COMPLEX REPRESENTATION OF A WAVE

Complex representation of a wave

Start with a wave propagating to the right:

$$y(x,t) = A\cos(kx - \omega t + \phi_0)$$

and write it in terms of the real part of a complex exponential:

$$y(x,t) = A\ \mathrm{Re}\left[e^{j(kx-\omega t+\phi_0)}\right]$$

The Re[] is usually suppressed:

$$\tilde{y}(x,t) = Ae^{j\phi_0}e^{j(kx-\omega t)}$$

Just like for the phasors, we define a complex amplitude

$$\tilde{A} = Ae^{j\phi_0}$$

and then simply write:

$$\tilde{y}(x,t) = \tilde{A}e^{j(kx-\omega t)}$$

The absolute value of $\tilde{A}$ gives the wave amplitude, and the phase of $\tilde{A}$ gives the initial phase ϕ_0.

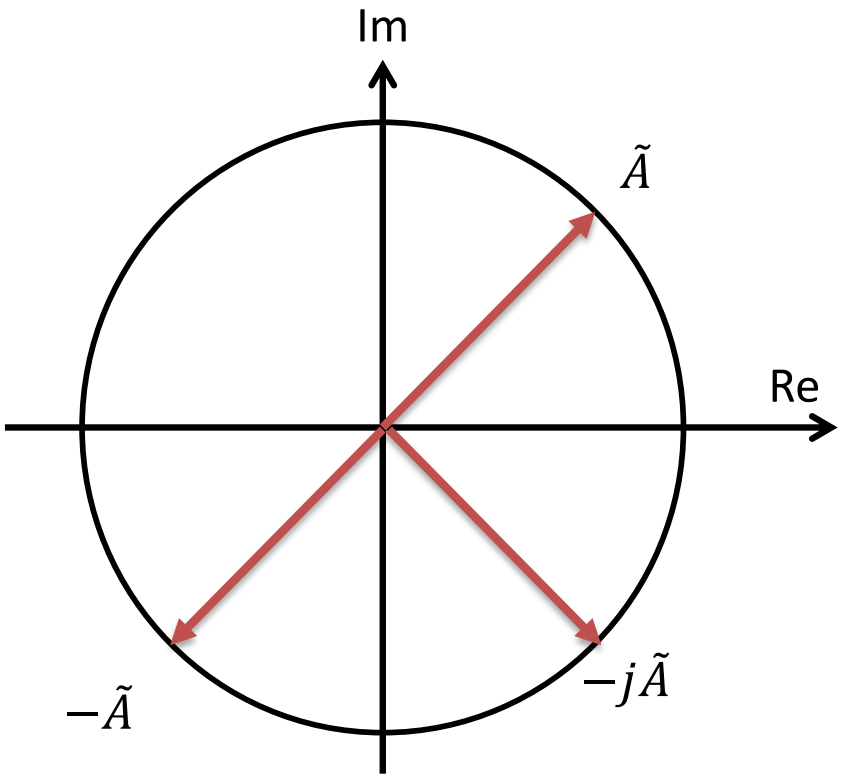
Complex representation of velocity and acceleration

$$\tilde{y}(x,t) = \tilde{A}e^{j(kx-\omega t)}$$

$$v_y = \frac{\partial \tilde{y}(x,t)}{\partial t} = -j\omega\tilde{A}e^{j(kx-\omega t)}$$

$$a_y = \frac{\partial^2 \tilde{y}(x,t)}{\partial t^2} = -\omega^2\tilde{A}e^{j(kx-\omega t)}$$

Where did that –j come from on $\tilde{A}$?



- Multiplication by -1 results in a 180° rotation in the complex plane
- Multiplication by –j results in a -90° rotation
- Differentiation leads to a -90° rotation
  - cos() → –sin()

Example

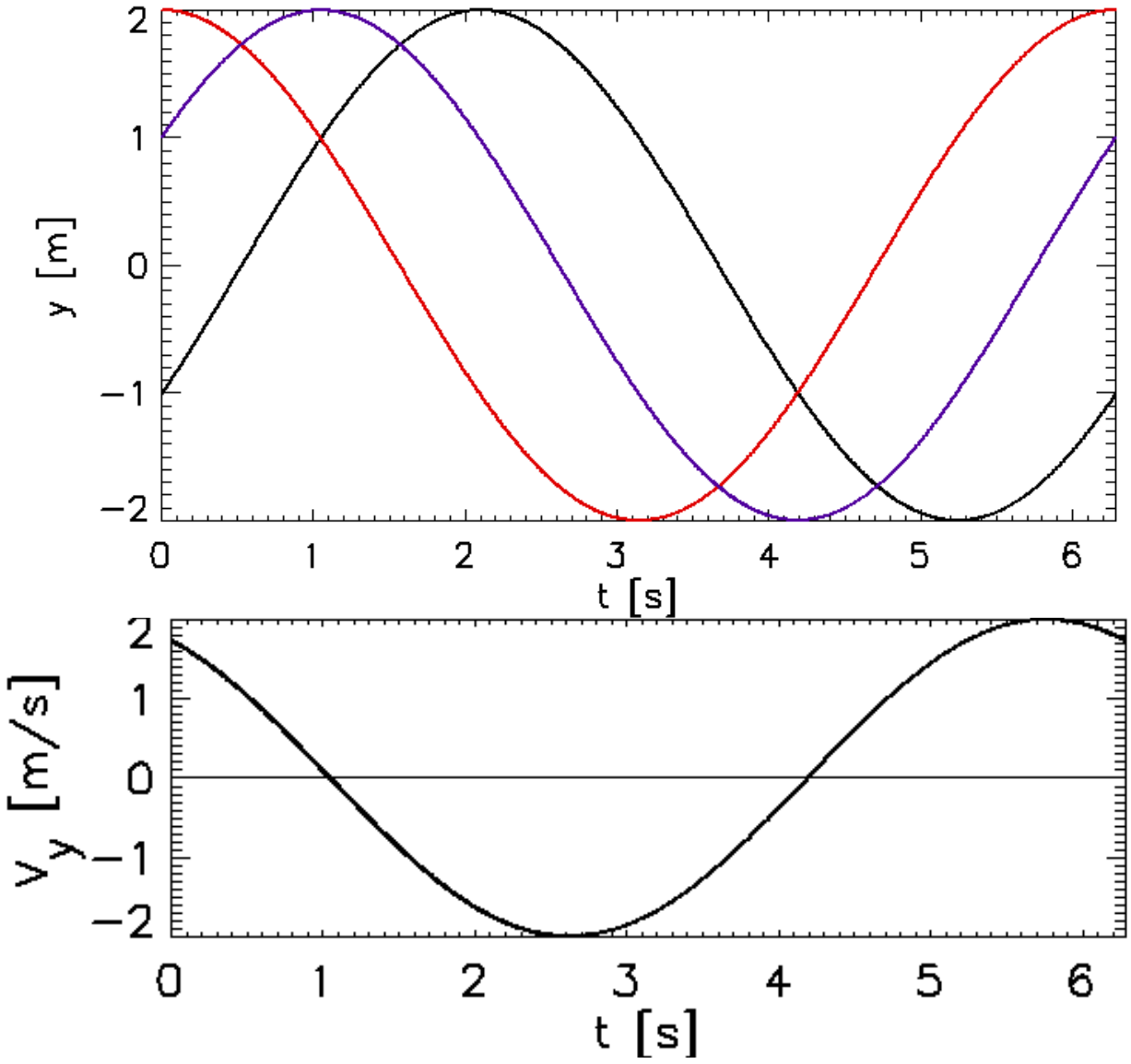
Two waves:

$$y_1(x,t) = (2.0\ \text{m})\sin(kx - \omega t + 3.67)$$
$$y_2(x,t) = (2.0\ \text{m})\cos(kx - \omega t)$$

travel together down a string with ω=1 rad/s.

Find the particle velocity (amplitude and phase) of the resulting superposition

$$y_3(x,t) = y_1 + y_2 = A\cos(kx - \omega t + \phi_0)$$



ex.3.1

y_1

y_2

$y_3 = y_1 + y_2$

Amplitudes all happen to be the same.

Summary

more info

- The medium does not move at the same speed as the wave. The medium velocity and acceleration are:

$$y = A\cos(kx - \omega t + \phi_0)$$

$$v_y = A\omega\sin(kx - \omega t + \phi_0)$$

$$a_y = -A\omega^2\cos(kx - \omega t + \phi_0)$$

- Wave equation: $\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$ with $v = \omega/k$
- Waves may be linearly superposed
- Waves can be written in complex notation as $\tilde{y}(x,t) = \tilde{A}e^{j(kx-\omega t)}$